



Value-driven energy prosumerism: Toward an integrated understanding of peer-to-peer energy trading adoption

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ABSTRACT

The growth of residential renewable energy has created new opportunities for decentralized energy systems, including peer-to-peer energy trading (P2P-ET). While the technical potential of P2P-ET is increasingly recognized, its broader adoption remains contingent on a complex set of behavioral drivers. This study proposes a two-tier behavioral framework to explain how individual value orientations (biospheric, egoistic, hedonic and altruistic) and experience with similar technologies influence adoption intentions through perceptual pathways. The proposed model was empirically tested using survey data from 456 participants. Multiple mediation analysis was conducted to examine both direct and mediated relationships among key constructs. Contrary to popular belief that the profit-making feature of P2P-ET is its biggest selling point, the results of our study show that participation intention was mainly predicted by hedonic motivation instead of utilitarian motivation. Warm glow effect, anticipated fun and sense of novelty contributed to stronger hedonic motivation. We observed positive total effect of biospheric and egoistic orientation, supporting the dual-appeal of P2P-ET model: environmental preservation and profit-making. Participants with similar experience showed greater willingness to participate in P2P-ET and perceived availability of facilitating conditions and the strength of social support more favourably. By integrating abstract value orientations and context-specific measures adapted from the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2), this research offers a systematic approach to disentangling the psychological and contextual pathways that shape adoption behaviour in P2P-ET, pathing the way to design more targeted outreach strategies to promote energy prosumerism.

1. Introduction

Peer-to-peer energy trading (P2P-ET) model has received more attention in recent years due to increase in ownership of distributed energy resources (DERs) among households and industries. Simply put, P2P-ET is an alternative energy trading management model that allows direct energy trading between energy prosumers (energy users that both produce and consumer energy) and consumers. P2P-ET is touted as an alternative energy management model that can boost grid resiliency and curtail network loss (Tushar et al., 2020; Saatloo et al., 2023), empower energy users (Wilkinson et al., 2020) and promote energy democracy (Dorahaki et al., 2023), increase cost savings (Neves et al., 2020) and self-sufficiency (Karami and Madlener, 2022), while promoting social trusts and community attachment (Fell, 2019). P2P-ET is not a novel idea, pilot projects were already being carried out in the United States as early as 2015 (Irena, 2020). However, real-world implementation of

P2P-ET remains very limited, hindered by high investment cost, the lack of supportive regulatory frameworks, complexity of the technology and pricing mechanism, as well as low public awareness and engagement among energy users (Gunarathna et al., 2022).

To address low public awareness and engagement, we need to develop a better understanding of the psychosocial aspect of P2P-ET. Existing literature on the psychosocial aspect of P2P-ET has predominantly focused on specific aspects of P2P-ET, with most attention being given to user preferences, pricing decisions and integration of blockchain (e.g. (Ecker et al., 2017), (Hahnel et al., 2020), (Pumphrey et al., 2020), (Amin et al., 2025b), (Amin et al., 2025c)). While these studies have enriched our understanding of how prosumers may behave in a P2P-ET environment, there remains a knowledge gap in how energy users make decisions related to whether they will take on the role of prosumers in P2P-ET. Filling this gap will further our understanding of how to effectively engage and onboard a broader range of energy users.

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To this date, most discussions of P2P-ET participation is built on the assumption of rational process that can be modelled using game theory and simulators (Soto et al., 2021). However, individuals' behaviour within the energy system is often not a simple product of rational choice. Instead, energy behaviours are inevitably influenced by cognitive biases, risk preferences and differences in subjective valuations of the tangible and intangible costs and benefits (Dorahaki et al., 2025). As such, participation in P2P-ET is likely to be an act shaped by both individual and contextual factors. However, the mechanisms through which these individual factors interact with contextual factors in shaping participation intention remains ambiguous in current literature.

To address this gap, we draw on theories within the broader domain of personality psychology. Dispositional traits are thought to be more effective in predicting spontaneous behaviours, while value construct is seen as a better predictor of planned action (Sagiv and Schwartz, 2022; Roccas et al., 2002). Traits describe enduring patterns of thoughts, emotions and behaviours (McCrae and Costa, 1997), in other words, they describe what people are like. On the other hand, values refer to cognitive representations of desirable goals that transcend situations, motivate action and serve as guiding principle in life (Schwartz, 2012). Values are seen as both normative and motivational (Hitlin and Piliavin, 2004). Drawing on De Groot and Steg's environmentally-significant value orientations (De Groot and Steg, 2008; Steg et al., 2014), four values may be of particular relevance within the context of P2P-ET: biospheric (concerns for the environment), altruistic (concern for others), egoistic (concern for personal resources) and hedonic (concern for pleasure and comfort) (Bouman et al., 2018). These values are deemed highly relevant here as they are well-aligned with the multiple appeals of P2P-ET: environmental preservation by promoting the use of renewable energies, improvement of local energy resiliency and citizen empowerment, monetization of energy surplus, as well as increased sense of novelty and fun in sustainable energy management. Since values may influence a person's attitude, perception, beliefs and behaviour (Schwartz, 2013; Schwartz and Zanna, 1992), variations in individual value orientation may determine how attractive the different appeals of P2P-ET are.

Values rarely exert direct influence on behaviour choice (Hitlin and Piliavin, 2004). Instead, values guide actions by influencing selection and interpretation of external stimuli, i.e. it affects our perceptual organizations (Ravlin and Meglino, 1987). It is thus expected that values affect behaviours, including decision-making, by influencing how we perceive various determinants of the said behaviour or decision, i.e., the relationships between values and behaviour are mediated by perceptual determinants. Since P2P-ET can be conceived as a technology-intensive innovation within energy management, we draw on constructs from the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) (Venkatesh et al., 2012a) to structure perceptual determinants of the behavioural intention (BI) to participate in P2P-ET. These perceptual determinants are performance expectancy (PE), effort expectancy (EE), social influence (SI), facilitating conditions (FC), hedonic motivation (HM), perceived price value (PV) and habits (HA) (Venkatesh et al., 2012a). Integrating the four environmentally significant value

orientations and the UTAUT2 model, we propose a two-tier motivational pathway to P2P-ET adoption (Fig. 1). This two-tier framework explicates the mechanism behind the dual appeal of P2P-ET, where adoption is driven by both "environmental preservation" and "self-interest".

Besides the role of value orientations, this study also sets out to examine the direct and indirect effects of experience with similar trading technologies on participation intention, as decision to adopt novel energy management technology like P2P-ET may also be affected by similar past experience. Exposure to similar technologies can influence how individual perceives new technology (Chen and Sintov, 2016). As illustrated in Fig. 1, the two-tier framework proposes that perceptual determinants mediate the effects of individual value orientations and experience on the intention to participate in P2P-ET.

The proposed conceptual framework of this research offers a more systematic and multifaceted approach to understanding and predicting the intention to participate in P2P-ET. Using survey data from 456 European energy users, this study demonstrates how enduring differences in abstract personal values translate into concrete decision to join/not join P2P-ET. Differing from existing research that often extrapolated findings from broader solar adoption research, this research offers to a more nuanced understanding of the P2P-ET model by tailoring our data collection and analysis to specifically focus on P2P-ET model. Furthermore, the insights generated from this study can help policymakers and stakeholders design more targeted policies and outreach strategies that align more effectively with individual motivations. In the next section, we explain the theories that underpin the proposed framework used in this study.

2. Theoretical background

Values affect our cognitions and emotions (Parks-Leduc et al., 2014). They influence preferences and decision-making by affecting perceived valences or attractiveness of different action alternatives (Feather, 1988, 1995, 2021), perceived risks and benefits (De Groot et al., 2013; Slimak and Dietz, 2006; Belanche et al., 2012) and perceived ethicality (Fritzsche and Oz, 2007; Pohling et al., 2016; Finegan and Theriault, 1997). Furthermore, as humans tend to strive for congruency between values, situation and choice of action, the degree of congruency between situations and a person's value orientation can elicit emotions that in turn influence behaviour and decision-making, e.g. through activation of heuristic in decision-making (Nelissen et al., 2007) (Quartz, 2009). Therefore, values provide an efficient and effective means to describe and explain similarities and differences across individuals, groups, nations, and cultures (de Groot and Steg, 2007).

Values only affect action/behaviour insofar they are cognitively activated due to context-relevance (Schwartz, 2012) and central to the self (Verplanken and Holland, 2002). As such, it is likely that not all values are equally salient within the context of P2P-ET. Drawing on Schwartz' value theory (Schwartz, 2012), P2P-ET is conceivably a business model that encompasses elements of the two opposing dimensions of the value continuum: self-transcendence (moving beyond selfish interests to contribute to the well-being of others and the

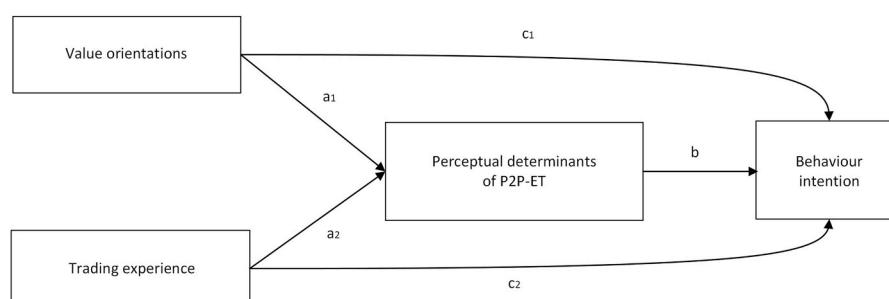


Fig. 1. Proposed relationship between values, experience, perceptions and behaviour intention of P2P-ET.

environment) and self-enhancement (the pursuit of one's own interests and gains). It can be seen as self-transcendence action in relation to its contribution to the promotion of DERs adoption, community energy resiliency, better grid balancing and support for egalitarianism. On the other hand, it is self-enhancing when individual/household is motivated by profit-maximization and gratification of sensation-seeking, personal status and control.

Within the self-transcendence and self-enhancement dimension, four core values are frequently used to explain environmentally-significant behaviours – biospheric, egoistic, altruistic and hedonic values (De Groot and Steg, 2008; Steg et al., 2014; Bouman et al., 2018; Choi and Johnson, 2019; Steg and De Groot, 2012). Biospheric and altruistic are inherently self-transcendence, while egoistic and hedonic values are self-enhancing (Steg et al., 2014; Bouman et al., 2018). In the studies related to pro-environmental behaviours and green technologies adoption, biospheric and altruistic values often predict stronger behaviour intention (Wolske et al., 2017; Rahardja et al., 2022; Fornara et al., 2016; Palm, 2020), while egoistic and hedonic value are usually associated with stronger resistance (Steg et al., 2014; Contzen et al., 2021a; Kastner and Matthies, 2016; Rezvani et al., 2018). Although P2P-ET cannot be fully construed as pro-environmental behaviour, we expect that these values to remain salient within P2P-ET context and can therefore, predict the intention to participate P2P-ET.

3. Methods

3.1. Study design and data collection

This research employed cross-sectional survey design to investigate the effect of personal value orientations and similar experience on behaviour intention mediated by perceptual determinants of P2P-ET (Fig. 2). Participants were recruited through online survey platform Prolific¹ and the survey was carried out via online platform Qualtrics.² Upon consenting to participate in the study, participants were prompted to provide information on demographic characteristics and whether they have performed any online trading beforehand (e.g., financial instruments, collectibles, etc.). Following that, participants were asked to rate 17 statements on their personal value orientation (using items from Environmental-Portrait Value Questionnaire (E-PVQ) (Bouman et al., 2018)). Thereafter, participants were presented with a short description and graphical illustration of P2P-ET, followed by 21 statements assessing various dimensions of P2P-ET (Supplemental Material A and A.1). Subsequently, participants responded to three statements designed to measure their intention to participate in P2P-ET in the future. Fig. 2 shows the flow of the survey.

3.2. Participants and data collection

This study uses survey data collected in Europe. Europe is an ideal setting for studying P2P-ET acceptance due to its progressive energy policies, high renewable energy adoption and growing emphasis on citizen participation in the energy system. The European Union (EU) actively promotes energy transition through initiatives like the Clean Energy for All Europeans package, which encourages prosumer participation in energy markets (Clean Energy for All Europeans, 2019). Additionally, many European countries have well-established smart grid infrastructure, making P2P-ET more feasible. Participants were informed of the nature and purpose of the study, as well as the payment rate prior to the survey.

Several criteria were used to screen for suitable participants: 1) between specified age range of 18–99 years old; 2) fluency in English; 3) active role in household financial decision-making; 4) currently residing

in Europe. This ensures that the insights generated in this study come from adults with some financial responsibilities, who are more likely to make informed decisions about household energy usage and investments. Screening for English fluency ensures that eligible participants whose first language is not English could comprehend survey questions accurately, thereby minimizing errors due to misinterpretations. In addition, to improve the representativeness of our data, we ensured a balanced recruitment of participants across different European regions. This allows us to capture diverse perspectives while accounting for differences in regional energy policies. Our sample selection criteria help to ensure the validity and relevance of our findings.

After data cleaning, the final dataset comprises of 456 respondents (59 % of the respondents identified themselves as males, 39 % of the respondents identified themselves as females and 2 % as non-binary). The average age of the respondents was 33.7 years old, and 75.9 % of them have obtained a bachelor's degree or higher. In terms of region, 21.7 % of the respondents were residing in Eastern Europe, 27.9 % in Western Europe, 31.6 % in Southern Europe and 18.6 % in Northern Europe. A majority of respondents (69.7 %) reside in households consisting of 2–4 individuals. More than half of respondents (58.3 %) perceive their household income as being on par with the national average in their country of residence. Among the respondents, 25 % of them own at least one form of DERs. Detailed summary of descriptive statistics can be found in Supplemental Material B. Data collection and processing were approved by SIKT (Norwegian Agency for Shared Services in Education and Research) and the Research Ethics Committee of the Department of Psychology at the University of Oslo (ref. 27316274). Data collection was completed in November 2024.

3.3. Measures

Personal value orientations were measured using E-PVQ (Bouman et al., 2018). When used within the context of our study, the scale has Cronbach's alpha of 0.82, indicating good internal consistency. To assess participants' cognitive and emotional perceptions of P2P-ET, they were first presented with a one-page description of the P2P-ET concept, followed by 21-item scale assessing how they perceive various aspects of P2P-ET (Supplemental Material A). The 21-item scale was developed for the purpose of this study based on adaptation of UTAUT2 (Venkatesh et al., 2012a). All UTAUT2 constructs were included except for HA, which was omitted due to the early diffusion stage of P2P-ET. While some studies have inferred HA from prior experience with related technologies, we refrained from doing so as HA is operationalized here as context-specific, cue-bound automated behaviour that resulted from learning (Venkatesh et al., 2012a; Limayem et al., 2007). Due to the distinctive characteristics of energy as the underlying "commodity" compared to typical online traded assets (e.g. stock, collectibles), inferring HA from user experience in conventional online trading platforms becomes theoretically and empirically less certain. A person may trade shares with someone living in another continent, while a person who sold their energy would have most likely sold it to one of the households within the local community. Accordingly, prior experience with analogous trading technologies is incorporated as a distinct construct within our model. Our measure of "trading experience" is therefore, intended to capture familiarity, not automaticity. An additional construct was also included to capture the profit-making feature of P2P-ET – Monetary Incentive (MI). While PV captures the cognitive trade-off between the perceived benefits of using a technology and the cost of using it (Venkatesh et al., 2012b), MI targets the centrality of the money-making feature of P2P-ET in participation intention. The Cronbach's alpha of the adapted UTAUT2 scale used in this study is 0.87. Confirmatory Factor Analysis (CFA) was also carried out to validate the adapted scale. The result shows the adapted UTAUT2 scale has reasonably good fit, with χ^2 (224) = 685.26, $p < .001$, CFI = 0.891, RMSEA = 0.067, and SRMR = 0.069 (see Supplemental Material A.2 for more details).

¹ <https://www.prolific.com/>.

² <https://www.qualtrics.com/>.

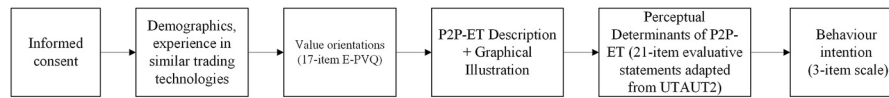


Fig. 2. Survey flow.

3.4. Statistical analyses

To investigate the underlying mechanisms linking value orientations and the intention to engage in P2P-ET, a multiple mediation model was specified and tested using *mediation* package in R (version 4.4.2). Specifically, the analysis tested whether value orientations and experience with similar trading technologies could predict BI to engage in P2P energy trading directly and indirectly through their effects on perceptual determinants (PE, EE, SI, FC, HM, PV and MI) of P2P-ET. Demographic variables (age, gender, education, household size, household income, ownership of DERs) were included as control variables to reduce potential confounding on BI. While some studies show that these factors are related to willingness to participate in P2P-ET and DERs adoption, they are not the primary focus of this study (see Supplemental Material E) for their estimate effects on BI. To assess the significance and sampling variability of the indirect effects, we employed nonparametric bootstrapping with 5000 resamples. This approach generated bias-corrected 95 % confidence intervals (CI) for all indirect paths. All key assumptions for mediation analysis were assessed before further analysis. Univariate and multivariate normality were tested. No extreme outliers or multicollinearity were detected. All analyses were done based on the unstandardized variables, and all coefficients reported in the Results section are unstandardized coefficients, which is customary in causal modelling (Hayes, 2009). To ensure the robustness of the results, we tested the model in *lavaan* using Structural Equation Modelling (SEM), treating all variables as observed. Both approaches yielded consistent estimates for the indirect effects, including the direction, magnitude, and statistical significance, supporting the stability of our findings across methodological frameworks.

4. Results

4.1. Direct and total effect of value orientation and trading experience on BI

To test the proposed multiple mediation model, we estimated a system of regression equations in which each mediator was regressed on the predictor set (Equation (1)), followed by an outcome model in which BI was regressed on all mediators, predictors and controls (Equation (2)).

Equation for M :

$$M_j = \alpha_j + \sum_i a_{ij} X_i + \varepsilon_{M_j} \quad (1)$$

Equation for $\hat{Y}$ (BI):

$$\hat{Y} = BI = \beta_0 + \sum_j b_j M_j + \sum_i c'_i X_i + \sum_r d_r C_r \quad (2)$$

$$\text{Total effect of } X_i \text{ on } \hat{Y} = c_i = c'_i + \sum_j a_{ij} b_j \quad (3)$$

Where X_i denotes the five predictors (biospheric, altruistic, egoistic, hedonic values, and experience), M_j denotes the seven mediators (PE, EE, SI, FC, HM, PV, MI), and C_r denotes control variables (age, gender, region, education, household size, household income, and DER ownership). Coefficients a_{ij} , b_j , c'_i , and c_i represent paths a , b , direct effects, and total effects, respectively.

Out of the four value orientations, as shown in Table 1, both biospheric and egoistic emerge as significant predictors of BI ($b = 0.2488$

Table 1

Total effects of predictors on BI (path c1 and c2), without mediators.

Predictor	Coefficient	SE
Biospheric	0.2488***	0.0463
Egoistic	0.1202**	0.0385
Hedonic	0.0544	0.0523
Altruistic	0.0285	0.0575
Experience	0.2402***	0.0682

Note: Unstandardized regression coefficients are reported. * $p < .05$, ** $p < .01$, *** $p < .001$.

and $b = 0.1202$ respectively). However, when the mediators were taken into consideration (Equation (3)), the direct effect of egoistic became non-significant ($b = 0.042$), indicating full mediation through mediators (Table 2) for egoistic value, and partial mediation for biospheric value. These findings suggest that P2P-ET is both directly and indirectly more attractive to individuals with stronger environmental concerns, whereas variation in BI among those with stronger egoistic orientations appears to arise primarily to the extent that egoistic values are associated with more favourable perceptions of P2P-ET. We also found that experience with similar directly and indirectly predicted BI ($b = 0.2402$ and $b = 0.1340$).

4.2. Indirect effects of value orientations and experience

The mediation analysis shows that many of the hypothesized indirect pathways are statistically significant (Table 3), indicating the presence of meaningful mediational mechanisms linking values, experience, perceptions and BI. Strongest mediated paths were observed through perceptual determinant HM. The extent to which participants anticipated enjoyment and pleasure from using P2P-ET was significantly influenced by their value orientation, whereby biospheric ($b = 0.0475$, $p < .001$), egoistic ($b = 0.0348$, $p < .001$) and hedonic values ($b = 0.0426$, $p = .003$) are associated with stronger HM. The result shows that the effect of hedonic value on BI is fully mediated by HM. Of the perceptual determinants, only PV cannot be explained by value orientations or similar experience, even though the perceived trade-off between the costs and benefits of P2P-ET was the second strongest perceptual determinant predicting BI (please refer to Supplemental Material C for details on path a1, a2 and b). Participants with similar experience appeared to perceive more social support ($b = 0.0197$, $p = .0140$) and better availability of necessary infrastructures ($b = 0.0304$, $p = .0140$).

BI = Behaviour Intention; PE = Performance Expectancy; EE = Effort Expectancy; FC = Facilitating Conditions; HM = Hedonic Motivation; SI = Social Influence.

Table 2

Direct effects of predictors on BI, with mediators.

Predictor	Coefficient	SE
Biospheric	0.1138**	0.0374
Egoistic	0.0419	0.0307
Hedonic	-0.0197	0.0412
Altruistic	-0.0235	0.0450
Experience	0.1340*	0.0542

Note: Unstandardized regression coefficients are reported. * $p < .05$, ** $p < .01$.

Table 3

Indirect effects (path a*b) of values and experience (significant effects only).

Path	Indirect Effect (95 % CI)	p
Biospheric -> PE -> BI	0.0131 [0.0005, 0.0332]	0.0368
Biospheric -> EE -> BI	0.0286 [0.0082, 0.0571]	0.0044
Biospheric -> FC -> BI	0.0191 [0.0033, 0.0411]	0.0140
Biospheric -> HM -> BI	0.0475 [0.0182, 0.0857]	0.0004
Egoistic -> PE -> BI	0.0097 [0.0001, 0.0239]	0.0464
Egoistic -> EE -> BI	0.0186 [0.0052, 0.0372]	0.0024
Egoistic -> HM -> BI	0.0348 [0.0130, 0.0611]	0.0004
Egoistic -> MI -> BI	-0.0088 [-0.0192, -0.0011]	0.0184
Hedonic -> HM -> BI	0.0426 [0.0144, 0.0782]	0.0028
Experience -> SI -> BI	0.0197 [0.0008, 0.0478]	0.0356
Experience -> FC -> BI	0.0304 [0.0054, 0.0637]	0.0140

Note. Unstandardized coefficients are reported. Confidence intervals are based on 5000 bootstrapped samples.

5. Discussion

5.1. Effects of value orientations

The main objective of the study is to uncover the extent to which differences in individual value orientation and exposure to similar trading technologies can directly and indirectly account for differences in the intention to participate in P2P-ET. The results show that when presented with the same description of P2P-ET, value orientations significantly explain the variances in the intention to participate in P2P-ET among European residential consumers and prosumers. Specifically, the extent to which participants prioritize concerns for the environment (biospheric) directly and indirectly predict participation intention. On the other hand, egoistic orientation did not directly predict intention. Instead, it is indirectly associated with intention through the hypothesized mediating pathways. The lack of direct path between egoistic orientation and intention suggests that egoistic orientation only meaningfully explains behaviour when it translates into favourable perceptions of P2P-ET. Additionally, the P2P-ET description (Supplemental Material A.1) provided in our survey might not have made egoistic goals (personal achievement and status) salient enough, which might in turn lead to weaker activation of egoistic orientation in comparison with biospheric orientation.

In line with our proposed mediating mechanism, we observed that stronger biospheric and egoistic orientation were associated with higher BI, with this association operating partly through more favourable perceptions of performance (PE) and ease of use (EE). Additionally, participants with stronger biospheric orientation also rated Facilitating Conditions (FC) higher, which can be interpreted as greater optimism regarding the availability of enabling infrastructures for P2P-ET. Together, these findings highlight that individuals whose value orientations are more congruent with the technology value propositions tend to have more positive views of technical enablers of the technology. One possible explanation for these observations is that when an action aligns with an individual's personal values, whether rooted in environmental concerns or personal gains, it feels more purposeful and worthwhile, which may in turn increase confidence in its effectiveness and the individual's ability to carry out the action. While these interpretations are theoretically grounded, it is important to note that our study did not directly investigate the psychological mechanisms linking personal values to efficacy beliefs. Future research should explore these underlying processes in greater depth.

Besides perceptions of technical enablers, technology adoption often hinges upon individual intrinsic motivation. Hedonic motivation (HM) has been regarded as an affective antecedent of technology adoption that serves as intrinsic motivator of behaviour (Tamilmani et al., 2019). HM is defined as the fun or pleasure derived from using a technology (Venkatesh et al., 2012a). Since P2P-ET is not a technology designed solely for fun or convenience purpose, we chose to include additional antecedent of pleasure –perceived environmental contribution. The idea

is that prosocial or pro-environmental behaviour can be partly self-rewarding through its *warm-glow effect* – the good feeling resulting from moral pride for being a “good person”, even if the actual benefits of the act is unclear (Andreoni, 1990). By participating in P2P-ET, individuals gain a private emotional benefit that adds to the pleasure aspect of P2P-ET, which in turn may motivate and encourage repeated engagement with P2P-ET. In this study, HM emerges as the primary determinant among the seven perceptual determinants included in our analysis.

Stronger orientation in biospheric, hedonic and egoistic in turn predicted stronger HM. The nuanced difference here is the specific aspect of P2P-ET that give rise to positive emotional reactions. Our findings show that higher BI due to higher HM attributable to perceived environmental contribution is observed among participants with stronger biospheric, as well as those with stronger hedonic drives. One plausible explanation is that participants with biospheric orientation experience stronger positive emotion through value congruence (Contzen et al., 2021b), since P2P-ET can be perceived as an act that is instrumentally meaningful for environmental preservation. At the same time, hedonically oriented participants see immediate positive-affect payoff due to warm-glow effect. Conversely, participants who are more driven by egoistic values are intrinsically motivated by anticipated fun and sense of novelty, but not by the perceived environmental contribution (see Supplemental Material D). In other words, moral pride did not motivate individuals with stronger egoistic orientation. Existing literature found that self-enhancement-oriented individuals may be especially responsive to self-focused positive emotions, such as pride (Tamir et al., 2016; Yin et al., 2024). Our findings highlight the distinction between *pride* (linked to achievement and status) and *moral pride* (grounded in ethical self-appraisal). There was no explicit contextual cue in our survey to activate achievement-based pride, i.e. we have not framed P2P-ET as an act that is personally impressive, competence-signalling or socially distinguishing. Taken together, this null effect suggests a boundary condition for prior findings, whereby pride motivates behaviour among egoistic-oriented individuals primarily when the pro-environmental act can be interpreted as personally advantageous or identity-enhancing, whereas morally framed pride alone may be insufficient to shift acceptance.

The financial aspect is often regarded as the key feature of the P2P-ET model. In this study, the financial aspect of P2P-ET is captured by PV and MI. PV is defined here as the perceived trade-off between costs and benefits of P2P-ET. Costs of using P2P-ET may include equipment investment, recurring platform fee, effort and time requirement, etc., while benefits of P2P-ET encompasses both tangible and intangible benefits, such as reduced electricity bills, better energy resiliency or satisfaction of being an active player in the energy system. On the other hand, MI focuses on the extent to which the profit-making feature of P2P-ET determines participation intention. A range of factors can affect the profitability of P2P-ET, including but not limited to energy price, tariff design, local energy generation and consumption patterns. We found that MI does not significantly predict BI, except among participants with strong egoistic orientation. Although P2P-ET is often touted for its profit-generating feature, our finding shows that even among participants who are egoistic-oriented, MI shows only a marginal association with BI.

Contrary to that, PV emerges as the second strongest perceptual determinant after HM. This finding is in line with existing literature that shows high cost remains as the main barrier of P2P-ET adoption (Junlakarn et al., 2022). Notably, we found no significant associations between any of the value orientations or experience and PV. It is likely that PV is largely dependent on the actual monetary costs (e.g. price of solar panels and battery), and less affected by subjective perceptions of other intangible costs and benefits, which are more sensitive to variances in value orientation and perceived familiarity. This result highlights the importance of contextual factors, such as investment costs and policies, in explaining the value-behaviour gap. Indeed, if contextual

factors are strongly supportive of the behaviour (e.g., investment cost in DERs is halved), public participation will increase regardless of individual value orientation (Steg et al., 2015).

Interestingly, no significant direct or indirect effect of altruistic values on BI was observed. While both biospheric and altruistic are categorized under self-transcendent values, altruistic pertains to concern for the welfare of other people, whereas biospheric pertains to concerns specific to the environment and nature itself (De Groot and Steg, 2008). In some instances, the two value orientations may conflict with each other, e.g. fossil fuel subsidies for the poor can lift the living standards of low-income communities but have adverse impacts on the environment. Our finding supports the view that biospheric is generally a better predictor than altruistic when the environmental aspect of an action is more salient than the prosocial aspect of it (Steg and De Groot, 2012). The absence of a significant effect of altruistic values on BI may also reflect a misalignment between altruistic motives and the saliency of individual benefits of P2P-ET, including the warm-glow appeal of P2P-ET. Warm-glow effect is seen as a form of *impure altruism* (Andreoni, 1990), where individuals are driven by self-image pay-off than the actual outcome of the public-good behaviour. Indeed, we found negative but not significant direct effect of altruistic, which may indicate altruistic individuals perceive misalignment between their core values and profit-associated activity like P2P-ET. Therefore, P2P energy sharing may appeal more to individuals with stronger altruistic orientation, as research shows that some households are willing to pay more for 100 % solar power and if the profit from energy generation is used for social cause (Cárdenas et al., 2022).

Sustainable innovations often struggle to gain public acceptance due to incongruence with one's value orientation (Contzen et al., 2021a). P2P-ET may be less susceptible to this type of public resistance, as it supports both self-enhancing and self-transcending values. Our study provides empirical support to the argument that P2P-ET can serve as an alternative way to incentivize DERs investment, even among individuals who are less inclined to act out of concerns for the environment. The results we obtain also highlight the importance of the frequently overlooked affective component of sustainable green technologies.

5.2. Cognitive biases in decision-making under uncertainty

The asymmetry between the effect of PV and the effect of MI on BI suggests that the decision to participate in P2P-ET is not driven solely by rational evaluation but may instead be partly shaped by cognitive biases. The presence of cognitive biases in decision making under uncertainty is highlighted in Kahneman and Tversky's Prospect Theory (Kahneman and Tversky, 1979), in which the authors argue that when faced with uncertainty, individuals evaluate potential outcomes as gains or losses relative to a reference point. Individuals also often deploy mental shortcuts or heuristics and unwittingly become victims of cognitive biases. In particular, the theory predicts that in most instances, losses loom larger than gains, and people tend to overweight outcomes that are considered certain, in comparison to outcomes that are probable, reflecting humans' tendency to succumb to loss aversion, certainty effect and status quo bias. In the context of energy research, failure to account for loss aversion leads to overestimation of DERs uptake (Knobloch et al., 2019) and may overestimate the effect of financial gain appeal of P2P-ET model. If each perceptual determinant is interpreted as a decision weight assigned by individuals when evaluating participation in P2P-ET, we may interpret the coefficients of PV and MI as the relative importance participants place on loss and gain. In light of this theoretical interpretation, we observe that participants placed approximately 1.76 times more weight on trade-off ($b_{pv} = 0.214$) than potential gain ($b_{mi} = -0.122$) of P2P-ET, somewhat consistent with existing findings that people, on average, experience losses as about twice as impactful as equivalent gains (e.g. (Tversky and Kahneman, 1992; Tom et al., 2007; Yechiam and Hochman, 2013)).

Besides loss aversion, the relative strength of PV in comparison with

MI may also reflect certainty effect. Certainty effect can be understood as the tendency to overweight outcomes they perceive as certain compared to outcomes that are merely probable. The costs of P2P-ET are considerably more certain than its potential profits, which may result in individuals assigning more decision-weight on the costs than the profits. In addition, this tendency to overweight the costs of P2P-ET may also be partly attributable to status quo bias, which refers to the tendency to stick with existing circumstances rather than adopting new alternatives, even when evidence suggests that change would lead to measurable improvements. P2P-ET challenges the notion of energy as something that is centrally produced and distributed. The current state is usually used as referent point in decision-making, and deviation from this referent point is perceived as either "gain" or "loss". If individuals live in a place with high energy supply security and relatively reasonable energy price, any change from the current state may quickly be seen as potential loss, especially when the status quo feels "endowed". Endowment effect tends to cause people to ascribe higher value to what they already "own" – the sense of stability from current energy management system. More often than not, these cognitive biases contribute to resistance to innovation. However, individuals with similar experience may be less susceptible to these cognitive biases as they are likely to perceive lower level of uncertainty in P2P-ET.

5.3. The role of prior similar experience as a key predictor

Our findings also highlight the importance of a key predictor that has often been overlooked: experience with similar technologies. We found that experience in using online trading platforms to trade financial instruments or collectibles directly and indirectly increases BI. In the original UTAUT2 model, prior experience is treated as a moderator, and the influence of antecedents such as PE and EE are expected to taper off as users become more familiar with the technology (Venkatesh et al., 2003). In our proposed framework, similar experience is thought to actively shape perceptions and beliefs related to P2P-ET instead of merely adjusting the importance of various antecedents of technology acceptance. In contrast to existing literature, where the effect of experience is typically confined to its direct influence on PE and EE (e.g. (Tavitiyaman et al., 2022; Taylor and Todd, 1995; Weerasinghe and Hindagolla, 2017)), our findings reveal a slightly different mechanism, whereby the strongest mediated effects were observed in the perceived availability of facilitating conditions (FC) and social supports (SI). Similar prior experience is associated with heightened users' awareness of institutional support systems (e.g., technical infrastructure, training programs), thereby strengthening the perception of FC. Higher SI among individuals with prior similar experience presumably reflect their tendency to engage with communities or peer groups invested in trading technologies or activities. This may form a self-reinforcing cycle, where experienced users gravitate toward others with shared interests, thus enhances their perception of social support and normative validation. The divergence we observe here implies a more dynamic, interpretative role of prior experience, where it enables prospective P2P-ET participants to recognize and leverage contextual and social resources. The positive role of similar experience found in this study highlights the importance of technical familiarity in enhancing situational confidence, i.e. individual's belief in their ability to effectively perform a specific behaviour or task within a given environmental context, based on their familiarity with the available resources, supports, and constraints of that situation. While trust and familiarity are conceptually distinct, familiarity underlies and enables trust formation (Luhmann, 2001), and trust is one of the fundamental elements of peer-to-peer business models (Hawlitschek et al., 2018).

6. Conclusion and limitations

This study sets out to better understand how differences in individual value orientations and prior experience with similar technologies

directly and indirectly influence the intention to adopt P2P-ET. The two-tier framework examined in this study reveals the underlying mechanisms of P2P-ET's dual appeal, demonstrating how both environmental concern and self-interest can motivate adoption. The mediated paths account for a substantial share of the total effect, demonstrating how broad ideals manifest into concrete technology appraisals, reconciling the gap between abstract psychology constructs and tangible aspects of energy technology. On their own, values often show low predictive power, because their influences tend to be indirect and context-dependent. In the context of P2P-ET, only biospheric orientation directly predict behavioural intention, while egoistic and hedonic values explain variances in intention through more proximal perceptions and emotions.

The findings of this study also show that while intrinsic motivation such as perceived fun and warm-glow effect is associated with stronger behavioural intention, lower participation intention was observed among participants who perceived negative trade-off between costs and benefits of P2P-ET. Meanwhile, the central feature of P2P-ET – profit-making, was not a substantive predictor of intention. Viewed through the lens of the Prospect Theory, whereby losses tend to loom larger than gains (Kahneman and Tversky, 1979), we observed a tendency for participants to place greater weight on potential losses (negative trade-off) than on potential profitability at a ratio of 1.76. Furthermore, the perceived trade-off between costs and benefits of P2P-ET appears to be independent of differences in participants' value orientation. Together, our findings underscore the need for deeper understanding of the role of cognitive biases in mental accounting, as well as the importance of contextual factors in decision-making.

While the novelty and distinctive features of P2P-ET prevents the integration of habit into our model, we integrated experience with similar technologies as a predictor in our theoretical framework. The results highlight the importance of similar experience, in particular, its role in strengthening situational confidence, observed through improved perceptions of infrastructure readiness and social supports. In turn, enhanced situational confidence may reduce the perception of uncertainty in decision-making and change the evaluative weighting individuals assign to potential barriers and enablers, thereby increasing the likelihood of P2P-ET adoption. Future research may investigate whether similar experience leads to lower perceived uncertainty and propensity for cognitive biases, as well as stronger trust. Understanding how similar experience shapes cognitive appraisals provide important insights into how exposure to similar technologies can be strategically leveraged, or how a lack of exposure can be addressed through alternative interventions to promote broader adoption.

There are two main limitations of this research. First, the research examined behaviour intention instead of actual behaviour. Changes in salient contextual factors, such as prices of DERs and electricity, may widen the gap between intention and actual behaviour. Consistent with situational strength theory (Cooper and Withey, 2009), we expect that strong contextual cues (e.g., large subsidies or zero-cost access) can restrict variances in behaviour and reduce the predictive power of individual differences. Second, the survey was conducted in Europe, where public awareness of DERs is generally high, and government policies are comparatively supportive of citizen-led energy initiatives. Moreover, since participants were recruited from online survey panel, the baseline technological aptitude and exposure to online trading platforms could be slightly higher than the general population. Nevertheless, the sample was considered broadly consistent with the typical profile of early adopters of P2P-ET (i.e., young, tech-savvy, relatively well-educated with modest income and basic knowledge of the energy system, residing in regions with more supportive energy policies). Future research may include samples with more diverse technological proficiency and environmental awareness to better capture the heterogeneity expected to be present in the early majority and late majority users of P2P-ET.

As causal inference remains debatable in cross-sectional study, future

research could also employ an experimental message-framing design to test whether different contextual cues selectively activate different value orientations and, in turn, shape behavioural intention. Comparing intentions across different conditions (e.g. community solidarity-focused versus pioneer status-focused) would help assess whether behaviour intention varies when value centrality changes. It would also clarify whether the weak or non-significant value effects observed here are better explained by limited cue saliency in our survey context rather than by the irrelevance of values for P2P-ET acceptance. In addition, while the study identifies parallel self-transcendent and self-enhancing appeals, future work could examine scenarios where these values conflict (e.g., green energy at much higher costs). If future P2P-ET pilots are implemented, researchers could examine whether behavioural intentions measured in surveys align with actual participation and trading behaviour in real-world settings. This would enable a direct test of the intention-behaviour gap and reveal whether values, differences in perceptions and emotions translate into observable engagement once households face real costs, real rewards, and real operational constraints.

7. Policy implications

To increase acceptance of P2P-ET, informational campaign must be framed strategically to balance between the self-transcending and self-enhancing features of P2P-ET. Gamification may increase the sense of novelty and joy, while enhancing user experience and engagement (Wang et al., 2025). However, research shows that while some households prefer price-setting in the form of gamification, others prefer the convenience of automated pricing (Ableitner et al., 2020). Providing users with the ability to select their preferred level of automation can serve as a critical design strategy to accommodate heterogeneous preferences and thereby foster wider adoption. In addition, communication strategy should address biases for loss and uncertainty aversion. Key contextual barriers – investment costs and technical complexity – continue to dominate decision-making as they increase the perceived tangible and intangible costs of P2P-ET, independent of individual factors. Another critical contextual factor that is frequently cited is the complexity of the market mechanism, which may also deter participation even among those who showed initial interest (Staudt and Richter, 2025). Domestic energy users are primarily concerned with the ease of trading process and payment (Pumphrey et al., 2020).

Besides the need to address contextual barriers, this research also underscores the untapped potential of leveraging the power of 'social proof' by strategically engaging individuals with similar experiences. Social proof is "a psychological phenomenon where people assume the actions of others are the correct behaviours to follow" ((Talib and Saat, 2017), p. 3). Individuals with experience in similar trading technologies represent a critical "low-hanging fruit" for accelerating adoption of P2P-ET. Their familiarity with similar interfaces, transaction mechanics, and risk-reward dynamics may reduce cognitive barriers to adoption. By strategically engaging these individuals as early adopters, they can enforce positive social influence, signals safety and reliability of P2P-ET, and normalize the use of P2P-ET. More research is needed to fine tune interdisciplinary approach to P2P-ET adoption such that policy, information campaigns and user interface can be synergistically designed to encourage participation and sustained engagement.

Research data statement

The dataset used in this study is not publicly available, as participants provided consent for their data to be used solely for the purposes of this research. However, the data may be made available by the corresponding author upon reasonable request and subject to ethical approval.

CRediT authorship contribution statement

Juliana Jia Yu Zhang: Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. **Knut Inge Fostervold:** Formal analysis, Methodology, Resources, Supervision, Writing – review & editing. **Anne Marie Halberg:** Methodology, Supervision, Validation, Writing – review & editing. **Cato Alexander Bjørkli:** Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clrc.2025.100380>.

Data availability

Data will be made available on request.

References

- Ableitner, L., Tiefenbeck, V., Meeuw, A., Wörner, A., Fleisch, E., Wortmann, F., 2020. User behavior in a real-world peer-to-peer electricity market. *Appl. Energy* 270, 115061.
- Amin, W., Huang, Q., Khan, A.A., Li, J., Afzal, M., 2025c. Risk-aware sellers and comfort-driven buyers: a game-theoretic P2P energy trading framework. *Smart Energy* 20, 100211.
- Amin, W., Huang, Q., Li, J., Khan, A.A., Subramaniam, U., Selvam, S., 2025b. A secure energy management model for peer-to-peer smart grids with user-centric constraints. *Internet of Things*, 101678.
- Andreoni, J., 1990. Impure altruism and donations to public goods: a theory of warm-glow giving. *The economic journal* 100 (401), 464–477.
- Belanche, D., Casaló, L.V., Flavián, C., 2012. Integrating trust and personal values into the technology acceptance model: the case of e-government services adoption. *Cuadernos de Economía y Dirección de la Empresa* 15 (4), 192–204.
- Bouman, T., Steg, L., Kiers, H.A.L., 2018. Measuring values in environmental research: a test of an environmental portrait value questionnaire. *Front. Psychol.* 9.
- Chen, B., Sintov, N., 2016. Bridging the gap between sustainable technology adoption and protecting natural resources: predicting intentions to adopt energy management technologies in California. *Energy Res. Social Sci.* 22, 210–223.
- Choi, D., Johnson, K.K., 2019. Influences of environmental and hedonic motivations on intention to purchase green products: an extension of the theory of planned behavior. *Sustain. Prod. Consum.* 18, 145–155.
- Clean energy for all Europeans package. <https://ec.europa.eu/energy/news/cep-update/>, 2019.
- Contzen, N., Handreke, A.V., Perlaviciute, G., Steg, L., 2021a. Emotions towards a mandatory adoption of renewable energy innovations: the role of psychological reactance and egoistic and biospheric values. *Energy Res. Social Sci.* 80, 102232.
- Contzen, N., Perlaviciute, G., Sadat-Razavi, P., Steg, L., 2021b. Emotions toward sustainable innovations: a matter of value congruence. *Front. Psychol.* 12, 661314.
- Cooper, W.H., Withey, M.J., 2009. The strong situation hypothesis. *Pers. Soc. Psychol. Rev.* 13 (1), 62–72.
- Cárdenas-Álvarez, J.P., España, J.M., Ortega, S., 2022. What is the value of peer-to-peer energy trading? A discrete choice experiment with residential electricity users in Colombia. *Energy Res. Social Sci.* 91, 102737.
- De Groot, J.I., Steg, L., Poortinga, W., 2013. Values, perceived risks and benefits, and acceptability of nuclear energy, risk analysis. *Int. J.* 33 (2), 307–317.
- de Groot, J.I.M., Steg, L., 2007. Value orientations to explain beliefs related to environmental significant behavior: how to measure egoistic, altruistic, and biospheric value orientations. *Environ. Behav.* 40 (3), 330–354.
- De Groot, J.I., Steg, L., 2008. Value orientations to explain beliefs related to environmental significant behavior: how to measure egoistic, altruistic, and biospheric value orientations. *Environ. Behav.* 40 (3), 330–354.
- Dorahaki, S., Muyeen, S.M., Amjadi, N., Qarnain, S.S., Benbouzid, M., 2025. Behavioral analytics for optimized self-scheduling in sustainable local multi-carrier energy systems: a prospect theory approach. *Sustainable Energy, Grids and Networks* 42, 101679.
- Dorahaki, S., Rashidinejad, M., MollahassaniPour, M., 2023. Overview of the peer-to-peer transactions and transactive energy concept, challenges, and outlook. In: Vahidinasab, V., Mohammadi-Ivatloo, B. (Eds.), *Demand-Side peer-to-peer Energy Trading*. Springer International Publishing, Cham, pp. 1–21.
- Ecker, F., Hahnel, U.J.J., Spada, H., 2017. Promoting decentralized sustainable energy systems in different supply scenarios: the role of autarky aspiration. *Front. Energy Res. S.*
- Feather, N.T., 1988. Values, valences, and course enrollment: testing the role of personal values within an expectancy-valence framework. *J. Educ. Psychol.* 80 (3), 381.
- Feather, N.T., 1995. Values, valences, and choice: the influences of values on the perceived attractiveness and choice of alternatives. *J. Personality Soc. Psychol.* 68 (6), 1135.
- Feather, N.T., 2021. Human values and the prediction of action: an expectancy-valence analysis. *Expectations and Actions*, pp. 263–289. Routledge.
- Fell, M.J., 2019. Social Impacts of peer-to-peer Energy Trading: a Rapid Realist Review Protocol. Centre for Research into Energy Demand Solutions (CREDS), Oxford, UK.
- Finegan, J., Theriault, C., 1997. The relationship between personal values and the perception of the corporation's code of ethics. *J. Appl. Soc. Psychol.* 27 (8), 708–724.
- Fornara, F., Pattitoni, P., Mura, M., Strazzeria, E., 2016. Predicting intention to improve household energy efficiency: the role of value-belief-norm theory, normative and informational influence, and specific attitude. *J. Environ. Psychol.* 45, 1–10.
- Fritzsche, D., Oz, E., 2007. Personal values' influence on the ethical dimension of decision making. *J. Bus. Ethics* 75, 335–343.
- Gunarathna, C.L., Yang, R.J., Jayasuriya, S., Wang, K., 2022. Reviewing global peer-to-peer distributed renewable energy trading projects. *Energy Res. Social Sci.* 89, 102655.
- Hahnel, U.J.J., Herberz, M., Pena-Bello, A., Parra, D., Brosch, T., 2020. Becoming prosumer: revealing trading preferences and decision-making strategies in peer-to-peer energy communities. *Energy Policy* 137, 111098.
- Hawliczek, F., Teubner, T., Gimpel, H., 2018. Consumer motives for peer-to-peer sharing. *J. Clean. Prod.* 204, 144–157.
- Hayes, A.F., 2009. Beyond baron and kenny: statistical mediation analysis in the new millennium. *Commun. Monogr.* 76 (4), 408–420.
- Hitlin, S., Piliavin, J.A., 2004. Values: reviving a dormant concept. *Annu. Rev. Sociol.* 30 (1), 359–393.
- Irena, T., 2020. Peer-To-Peer Electricity Trading Innovation Landscape Brief. IRENA, Masdar City, Abu Dhabi.
- Junlakarn, S., Kokchang, P., Audomvongseeree, K., 2022. Drivers and challenges of peer-to-peer energy trading development in Thailand. *Energies* 15 (3), 1229.
- Kahneman, D., Tversky, A., 1979. Prospect theory: an analysis of decision under risk. *Econometrica* 47 (2), 263–291.
- Karami, M., Madlener, R., 2022. Business models for peer-to-peer energy trading in Germany based on households' beliefs and preferences. *Appl. Energy* 306, 118053.
- Kastner, I., Matthies, E., 2016. Investments in renewable energies by German households: a matter of economics, social influences and ecological concern? *Energy Res. Social Sci.* 17, 1–9.
- Knobloch, F., Huijbregts, M.A.J., Mercure, J.-F., 2019. Modelling the effectiveness of climate policies: how important is loss aversion by consumers? *Renew. Sustain. Energy Rev.* 116, 109419.
- Limayem, M., Hirt, S.G., Cheung, C.M., 2007. How habit limits the predictive power of intention: the case of information systems continuance. *MIS Q.* 705–737.
- Luhmann, N., 2001. Confiance et familiarité: problèmes et alternatives. *Réseaux (Centre national d'études des télécommunications (France))* 108 (4), 15.
- McCrae, R.R., Costa Jr., P.T., 1997. Personality trait structure as a human universal. *Am. Psychol.* 52 (5), 509.
- Nelissen, R.M.A., Dijkster, A.J.M., de Vries, N.K., 2007. Emotions and goals: assessing relations between values and emotions. *Cognit. Emot.* 21 (4), 902–911.
- Neves, D., Scott, I., Silva, C.A., 2020. Peer-to-peer energy trading potential: an assessment for the residential sector under different technology and tariff availabilities. *Energy* 205, 118023.
- Palm, A., 2020. Early adopters and their motives: differences between earlier and later adopters of residential solar photovoltaics. *Renew. Sustain. Energy Rev.* 133, 110142.
- Parks-Leduc, L., Feldman, G., Bardi, A., 2014. Personality traits and personal values: a meta-analysis. *Pers. Soc. Psychol. Rev.* 19 (1), 3–29.
- Pohling, R., Bzdok, D., Eigenstetter, M., Stumpf, S., Strobel, A., 2016. What is ethical competence? The role of empathy, personal values, and the five-factor model of personality in ethical decision-making. *J. Bus. Ethics* 137, 449–474.
- Pumphrey, K., Walker, S.L., Andoni, M., Robu, V., 2020. Green hope or red herring? Examining consumer perceptions of peer-to-peer energy trading in the United Kingdom. *Energy Res. Social Sci.* 68, 101603.
- Quartz, S.R., 2009. Reason, emotion and decision-making: risk and reward computation with feeling. *Trends Cognit. Sci.* 13 (5), 209–215.
- Rahardja, F.A., Chen, S.-C., Rahardja, U., 2022. Review of behavioral psychology in transition to solar photovoltaics for low-income individuals. *Sustainability* 14 (3), 1537.
- Ravlin, E.C., Meglino, B.M., 1987. Effect of values on perception and decision making: a study of alternative work values measures. *J. Appl. Psychol.* 72 (4), 666.
- Rezvani, Z., Jansson, J., Bengtsson, M., 2018. Consumer motivations for sustainable consumption: the interaction of gain, normative and hedonic motivations on electric vehicle adoption. *Bus. Strat. Environ.* 27 (8), 1272–1283.
- Roccas, S., Sagiv, L., Schwartz, S.H., Knafo, A., 2002. The big five personality factors and personal values. *Pers. Soc. Psychol. Bull.* 28 (6), 789–801.
- Saatloo, A.M., Mirzaei, M.A., Mohammadi-Ivatloo, B., 2023. A robust decentralized peer-to-peer energy trading in community of flexible microgrids. *IEEE Syst. J.* 17 (1), 640–651.
- Sagiv, L., Schwartz, S.H., 2022. Personal values across cultures. *Annu. Rev. Psychol.* 73 (1), 517–546.
- Schwartz, S.H., 2012. An overview of the Schwartz theory of basic values. *Online readings in Psychology and Culture* 2 (1), 11.

- Schwartz, S., 2013. Value Priorities and Behavior: Applying a Theory of Integrated Value Systems, the Psychology of Values. Psychology Press, pp. 1–24.
- Schwartz, S.H., 1992. Universals in the content and structure of values: theoretical advances and empirical tests in 20 countries. In: Zanna, M.P. (Ed.), *Advances in Experimental Social Psychology*. Academic Press, pp. 1–65.
- Slimak, M.W., Dietz, T., 2006. Personal values, beliefs, and ecological risk perception. *Risk Anal.* 26 (6), 1689–1705.
- Soto, E.A., Bosman, L.B., Wollega, E., Leon-Salas, W.D., 2021. Peer-to-peer energy trading: a review of the literature. *Appl. Energy* 283, 116268.
- Staudt, P., Richter, B., 2025. Empirical case study of a digitally enabled energy community with prosumers and P2P trading. *Energy, Sustainability and Society* 15 (1), 6.
- Steg, L., Perlaviciute, G., Van der Werff, E., 2015. Understanding the human dimensions of a sustainable energy transition. *Front. Psychol.* 6, 805.
- Steg, L., Perlaviciute, G., Van der Werff, E., Lurvink, J., 2014. The significance of hedonic values for environmentally relevant attitudes, preferences, and actions. *Environ. Behav.* 46 (2), 163–192.
- Steg, L., De Groot, J.I., 2012. Environmental Values, the Oxford Handbook of Environmental and Conservation Psychology. Oxford University Press, pp. 81–92.
- Talib, Y.Y.A., Saat, R.M., 2017. Social Proof in Social Media Shopping: an Experimental Design Research, SHS Web of Conferences. EDP Sciences, 02005.
- Tamilmani, K., Rana, N.P., Prakasam, N., Dwivedi, Y.K., 2019. The battle of brain vs. heart: a literature review and meta-analysis of “hedonic motivation” use in UTAUT2. *Int. J. Inf. Manag.* 46, 222–235.
- Tamir, M., Schwartz, S.H., Cieciuch, J., Riediger, M., Torres, C., Scollon, C., Dzokoto, V., Zhou, X., Vishkin, A., 2016. Desired emotions across cultures: a value-based account. *J. Personality Soc. Psychol.* 111 (1), 67.
- Tavitiyaman, P., Zhang, X., Tsang, W.Y., 2022. How tourists perceive the usefulness of technology adoption in hotels: interaction effect of past experience and education level. *J. China Tourism Res.* 18 (1), 64–87.
- Taylor, S., Todd, P., 1995. Assessing IT usage: the role of prior experience. *MIS Q.* 561–570.
- Tom, S.M., Fox, C.R., Trepel, C., Poldrack, R.A., 2007. The neural basis of loss aversion in decision-making under risk. *Science* 315 (5811), 515–518.
- Tushar, W., Saha, T.K., Yuen, C., Smith, D., Poor, H.V., 2020. Peer-to-Peer trading in electricity networks: an overview. *IEEE Trans. Smart Grid* 11 (4), 3185–3200.
- Tversky, A., Kahneman, D., 1992. Advances in prospect theory: cumulative representation of uncertainty. *J. Risk Uncertain.* 5, 297–323.
- Venkatesh, V., Morris, M.G., Davis, G.B., Davis, F.D., 2003. User acceptance of information technology: toward a unified view. *MIS Q.* 425–478.
- Venkatesh, V., Thong, J.Y., Xu, X., 2012b. Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. *MIS Q.* 157–178.
- Venkatesh, V., Thong, J.Y.L., Xu, X., 2012a. Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. *MIS Q.* 36 (1), 157–178.
- Verplanken, B., Holland, R.W., 2002. Motivated decision making: effects of activation and self-centrality of values on choices and behavior. *J. Pers. Soc. Psychol.* 82 (3), 434–447.
- Wang, Y., Gu, C., Xie, D., Alhazmi, M., Kim, J., Wang, X., 2025. Enhancing peer-to-peer energy trading in integrated energy systems: gamified engagement strategies and differentiable robust optimization. *Energy Rep.* 13, 3225–3236.
- Weerasinghe, S., Hindagolla, M., 2017. Technology acceptance model in the domains of LIS and education: a review of selected literature. *Library Philosophy and Practice*, pp. 1–27.
- Wilkinson, S., Hojckova, K., Eon, C., Morrison, G.M., Sandén, B., 2020. Is peer-to-peer electricity trading empowering users? Evidence on motivations and roles in a prosumer business model trial in Australia. *Energy Res. Social Sci.* 66, 101500.
- Wolske, K.S., Stern, P.C., Dietz, T., 2017. Explaining interest in adopting residential solar photovoltaic systems in the United States: toward an integration of behavioral theories. *Energy Res. Social Sci.* 25, 134–151.
- Yechiam, E., Hochman, G., 2013. Losses as modulators of attention: review and analysis of the unique effects of losses over gains. *Psychol. Bull.* 139 (2), 497.
- Yin, S., Wang, Y., Liu, Y., Wang, S., 2024. Exploring drivers of behavioral willingness to use clean energy to reduce environmental emissions in rural China: an extension of the UTAUT2 model. *J. Renew. Sustain. Energy* 16 (4).